RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi M 251801167-2025-AIDS

M2

Started on	Thursday, 24 September 2026, 10:25 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:46 AM
Time taken	20 mins 49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++){
        int c=a[i];
        for(int j=i+1;j<n;j++){
            if(c==a[j])
            {
               printf("%d",a[j]);
                break;
            }
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi M 251801167-2025-AIDS

M2

Started on	Thursday, 24 September 2026, 10:56 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:57 AM
Time taken	44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int sum = 0;
    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        sum += a[i];
    }

    int expected = n * (n - 1) / 2;

    printf("%d", sum - expected);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi M 251801167-2025-AIDS

M2

Started on	Thursday, 24 September 2026, 10:59 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:14 AM
Time taken	15 mins 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int T;
    scanf("%d", &T);
    while (T--) {
        int n;
        scanf("%d", &n);
        int a[n];
        for (int i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }
        int m;
        scanf("%d", &m);
        int b[m];
        for (int i = 0; i < m; i++) {
            scanf("%d", &b[i]);
        }
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                if (a[i] == b[j]) {
                    printf("%d ", a[i]);
                    break;
```

```
                    break;
                }
            }
        }
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi M 251801167-2025-AIDS

M2

Started on	Thursday, 24 September 2026, 11:15 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:20 AM
Time taken	4 mins 17 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int T;
    scanf("%d", &T);
    while (T--) {
        int n;
        scanf("%d", &n);
        int a[n];
        for (int i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }
        int m;
        scanf("%d", &m);
        int b[m];
        for (int i = 0; i < m; i++) {
            scanf("%d", &b[i]);
        }
        int i = 0;
        int j = 0;
        while (i < n && j < m) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
```

```
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }
        printf("\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi M 251801167-2025-AIDS

M2

Started on	Thursday, 24 September 2026, 11:20 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:26 AM
Time taken	5 mins 35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int m,ans=0;
    scanf("%d",&m);
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(abs(a[i]-a[j])==m){
                ans=1;
            }
        }
    }
    printf("%d",ans);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔

	Input	Expected	Got	
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi M 251801167-2025-AIDS

M2

Started on	Thursday, 24 September 2026, 11:27 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:38 AM
Time taken	11 mins 12 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
int main() {
    int n;
    scanf("%d", &n);
    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    int m;
    scanf("%d", &m);
    int i = 0;
    int j = 1;
    int ans = 0;
    while (i < n && j < n) {
        if (i == j) {
            j++;
        }
        else if (abs(a[i] - a[j]) == m) {
            ans = 1;
            break;
        }
        else if (abs(a[i] - a[j]) < m) {
            j++;
        }
        else {
            i++;
        }
    }
    printf("%d", ans);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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